

F-TALENT CODE 2026

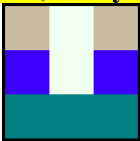
Canvas Restoration

Time limit: 1.00 s | Memory limit: 512 MB

Problem Statement

Mondrianism is a popular art movement originating from the city of Pixelburg. Paintings in this style strictly use a palette of only three colors: Honeydew, Indigo, and Teal. The canvas itself is structured as a grid of rows and columns. To create a authentic piece, an artist starts with an empty canvas and applies a series of brushstrokes. Every single brushstroke must completely cover either an entire row or an entire column with one solid color from the palette, overwriting any colors previously painted on those cells. A shady merchant in the back alleys of Pixelburg wants to sell you a painting that he claims to be a genuine Mondrianist artwork.

Given the final color configuration of the canvas, can you figure out if his claim is true?



Input

The first line of the input has two integers n and m , describing the height and the width of the painting.
The following n lines have strings of length m , describing each row of the canvas. The colors Honeydew, Indigo, and Teal are denoted by 'H', 'I', and 'T', and an unpainted part of a canvas by '.'.

Output

Output a sequence of brushstrokes that would produce the given painting or state that it is impossible by outputting "Impossible". If such a sequence of brushstrokes exists, first write the number of brushstrokes and then three values for each brushstroke: whether we paint a row ('R') or a column ('C'), its index, and the color of the brushstroke.

Constraints

$1 \leq n, m \leq 1000$
You may use at most 10^5 brushstrokes.

Example 1:

INPUT	OUTPUT
3 3 .H. IHI TTT	3 R 2 I C 2 H R 3 T